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Parsa Majidi

Embedded Systems Engineer

Linkedin
github

Specializing in AI acceleration, firmware development, and low-level optimization for embedded Android and Linux platforms. Currently contributing at Kynetics, focusing on BSP integration, driver development, and performance tuning. Pursuing an M.Sc. in ICT for Internet and Multimedia – Smart Industry 4.0, with a research focus on embedded AI and real-time computing. Passionate about optimizing embedded architectures and pushing the boundaries of AI-driven system design.

Education

University of Padova, Italy | Supervisor: Prof. Andrea Triossi — **Oct 2022 – October 2025**

M.S. ICT for Internet and the Internet - Smart Industry 4.0

- **Thesis:** Implementation of Tree Tensor Networks, a quantum-inspired machine learning algorithm, on the AI Engine of Versal adaptive SoCs for ultra-low latency applications
  - Vitis, Intrinsics APIs, AIE APIs, Quantum Machine Learning Algorithms, Versal Architecture, Multiply-Accumulate

University of Mazandaran, Iran — **Oct 2018 – July 2022**

B.S. Electrical Engineering, GPA: 3.66/4.00

- **Thesis:** Driver Drowsiness Detection | Tensorflow lite, OpenCV, Image Processing, Raspberrypi, TinyML - Grade: 20/20

Experience

Kynetics Inc. - Intern → Part-time → Apprentice — **Sep 2023 – Present**

Firmware | Embedded System Engineer | C/C++, Android/Linux BSPs, Firmware, Kernel, OTA — Padova, IT

- Developed and integrated USB Power Delivery (PD) firmware in C, increasing USB-C hub compatibility from 40% to 80%
- Developed and ported Android (AOSP) and Linux BSPs for custom hardware platforms
- Implemented and debugged kernel device drivers (Ethernet, I2C, CAN, Bluetooth)
- Utilized tools such as logic analyzers, oscilloscopes, and JTAG debuggers for low-level debugging
- Contributed to the spi-tools open-source project — Commit
- Built demo Android applications to test firmware and hardware integration

Projects

Porting and Developing Android 14 on Verdin iMX8MP

- Adapted and optimized Android 14 for the Verdin i.MX 8M Plus module
- Achieved 70–90% boot-time reduction by enabling and fine-tuning hibernation features
- Customized the Linux kernel, optimized bootloader settings, and integrated peripheral support

Neural Network Implementation on FPGA Aritx A7 — **Link**

- Implementation of a Keras neural network using HLS4ML on an Artix-7 FPGA
- Integrated with a custom VHDL-based UART module for real-time serial communication

Programming the TurtleBot for Following a Path

- Team project for a line-following robot using STM32 microcontroller
- Ranked 1st out of 10 teams in Embedded Real-Time Control course by optimizing PID control for fastest line-tracking

Skills

Programming	C/C++, Python, Verilog, VHDL, Rust, Java, CUDA
Embedded	Embedded Linux, Android/Linux BSP, FPGA, Yocto, ARM, QEMU, PCB
Debugging	JTAG, GDB, Logic Analyzer
Protocols	Ethernet, I2C, UART, SPI, CAN
Software	Vitis Unified IDE, Vivado, Altium Designer
Tools	Docker, Git, Jira, Bitbucket, Github

Language

English	Professional Proficiency
Persian	Native
Italian	Intermediate